Reference Notes

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Here is a Python script that sequences your requested actions. To ensure the script actually reaches the end without terminating at the first intentional mistake, the deliberate crashes are wrapped in `try...except` blocks so you can see the exact errors they generate while allowing the code to continue.

```
import matplotlib.pyplot as plt
import numpy as np

# 1. Print Mistake (Deliberate Crash)
print("--- Step 1: Print Mistake ---")
try:
    # Attempting to concatenate a string with an integer without casting
    print("The score is: " + 100)
except Exception as e:
    print(f"CRASH 1: {type(e).__name__} - {e}")

# 2. First Plot (Success)
print("\n--- Step 2: First Plot ---")
plt.figure(1)
plt.plot([1, 2, 3], [2, 4, 6], color='red', marker='x')
plt.title("Successful Line Plot")

# 3. plt Mistake (Deliberate Crash)
print("\n--- Step 3: plt Mistake ---")
try:
    # Attempting to plot arrays of different lengths
    plt.plot([1, 2, 3, 4], [10, 20])
except Exception as e:
    print(f"CRASH 2: {type(e).__name__} - {e}")

# 4. Two Print Successes
print("\n--- Step 4: Print Successes ---")
print("Successfully caught the dimension mismatch error.")
print("Proceeding to generate the final image grid...")

# 5. Final Plot with imshow and savefig (Success)
plt.figure(2)
# Create a 10x10 grid of random values for the image
```

```
image_matrix = np.random.rand(10, 10)

# Display data as an image
plt.imshow(image_matrix, cmap='plasma', interpolation='nearest')
plt.colorbar(label="Intensity")
plt.title("2D Array Imshow Plot")

# Save the figure to your local directory before showing it
plt.savefig("test_output_image.png", dpi=300, bbox_inches='tight')

# Render all open figures to the screen
plt.show()
```

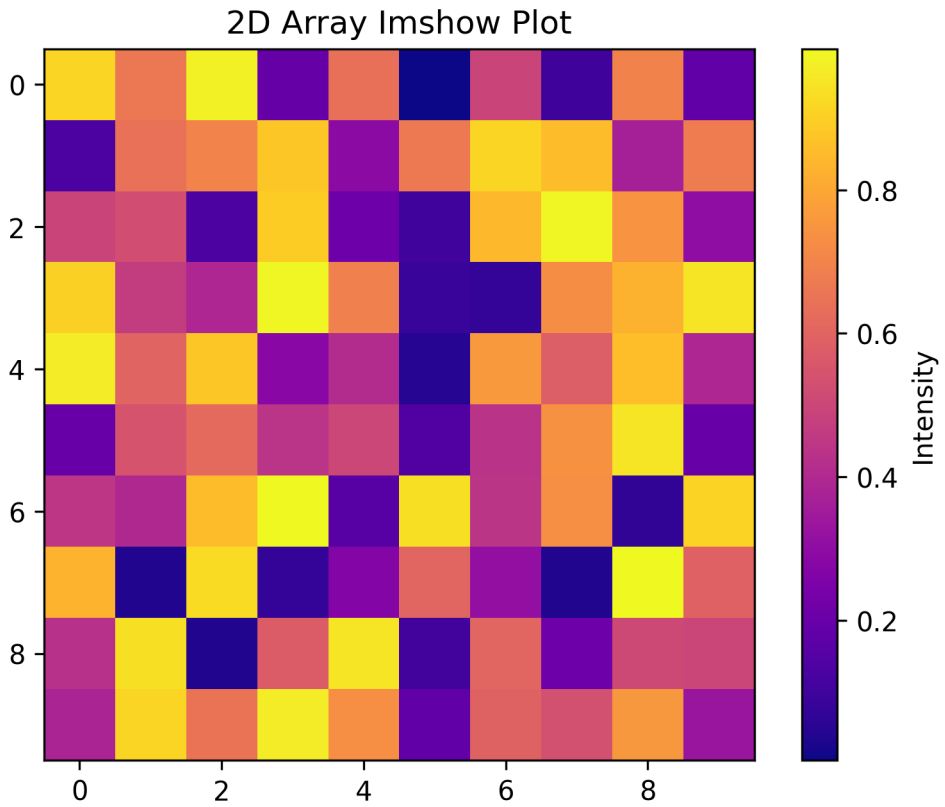
OUTPUT:

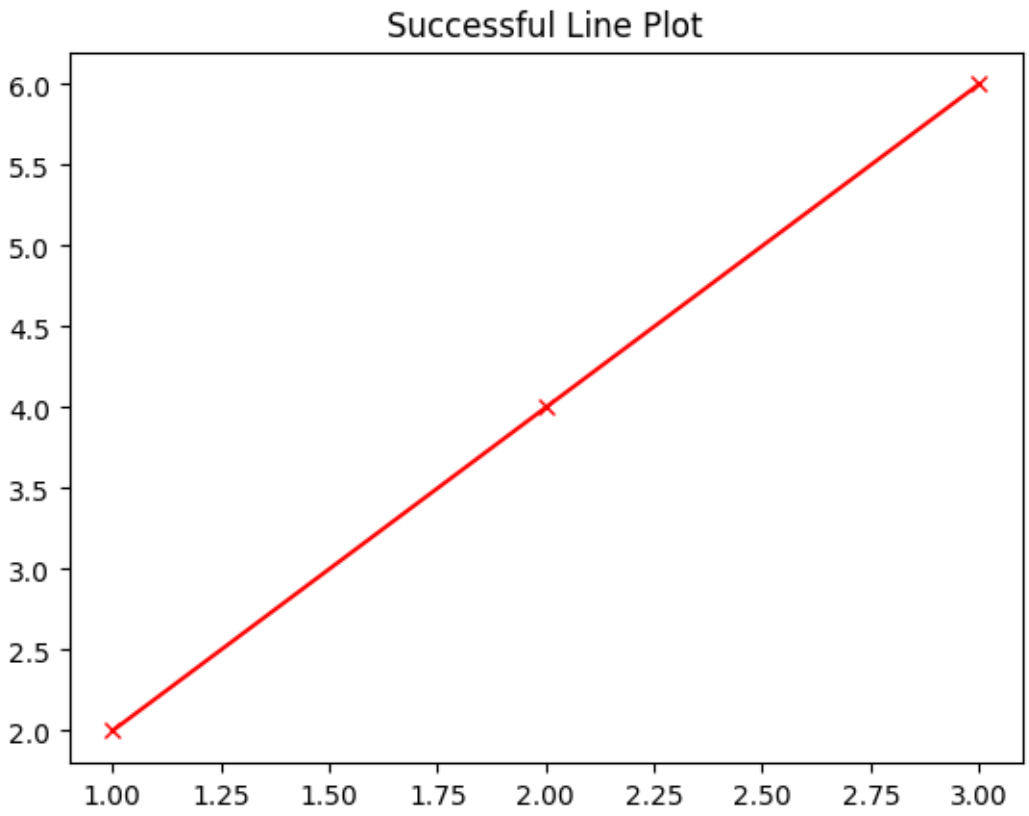
```
--- Step 1: Print Mistake ---
CRASH 1: TypeError - can only concatenate str (not "int") to str

--- Step 2: First Plot ---

--- Step 3: plt Mistake ---
CRASH 2: ValueError - x and y must have same first dimension, but have shapes
↪  (4,) and (2,)

--- Step 4: Print Successes ---
Successfully caught the dimension mismatch error.
Proceeding to generate the final image grid...
```





Code Explanation

1. **Print Mistake (`TypeError`):** Python is strongly typed. Attempting to use the `+` operator to combine a string (`"The score is: "`) and an integer (`100`) forces a hard crash. You must explicitly cast the integer using `str(100)` or use an f-string (`f"The score is: {100}"`).
2. **First Plot:** A standard line plot is generated successfully on `Figure 1` because the X and Y lists have the exact same number of elements (3).
3. **plt Mistake (`ValueError`):** The `plt.plot()` function requires the `x` and `y` arrays to have identical dimensions. Passing an array of 4 items for X and 2 items for Y causes matplotlib to crash because it cannot map the coordinates.
4. **Print Successes:** Standard, error-free string prints execute normally.
5. **Final Plot (`imshow` and `savefig`):**

- `plt.imshow()` takes a 2D array (like a matrix or a loaded image file) and maps the numerical values to colors using a colormap (`cmap='plasma'`).
- `plt.savefig("filename.png")` writes the current active figure to your hard drive. This must be called *before* `plt.show()`, because `show()` often clears the figure buffer from memory once the window is closed, which would result in saving a blank image.